ANSWERS.

Year 10

Pathway 2

Test 5

2019

Baldivis
Secondary College

PROBABILITY

Name: ____________________ Class: ______ Time: 60 minutes Total Mark:____/55

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning will be allocated zero marks.

Resources: 1 A4 page of notes (1 side), Calculator allowed.

Question 1

3 marks (1, 1, 1)

A bag contains 20 balls. Six are green, nine are blue and the rest are white. If one ball is chosen at random from the bag, find the probability that it is

a. Green

$\frac{6}{20} = \frac{3}{10}$

b. White

$\frac{5}{20} = \frac{1}{4}$

c. Blue

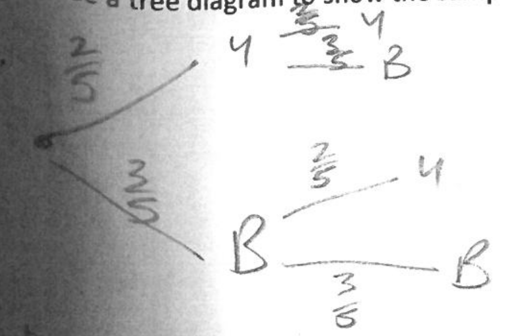
$\frac{9}{20}$

Question 2

5 marks(3, 2)

A bag contains 2 yellow and 3 black marbles. Two marbles are drawn at random, with replacement.

a. Use a tree diagram to show the sample space



YY
YB
BY
BB.

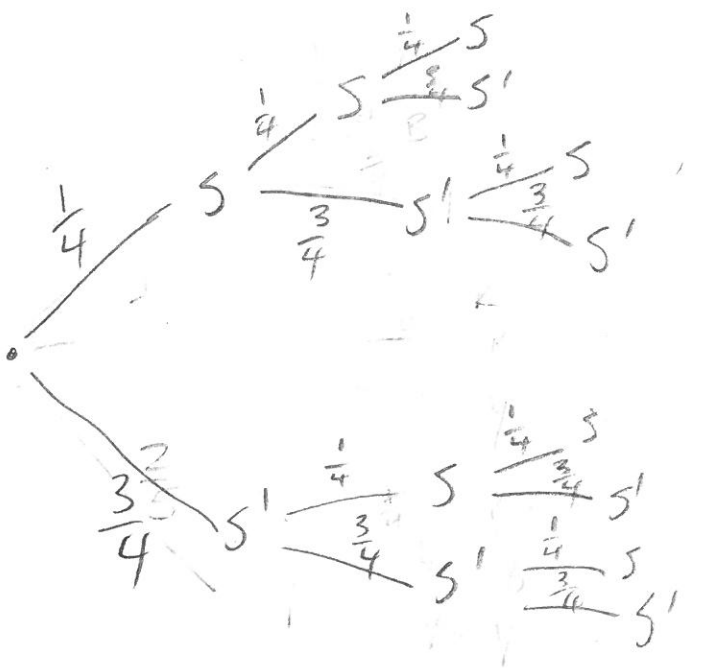
b. Find the probability that the marbles are both yellow

$\frac{4}{25}$

Question 3	8 marks (4, 2, 2)

A card is cut from a normal 52-card pack and the suit is noted. The card is replaced and the process is twice repeated.

a) Draw a simplified (weighted) tree diagram to show the sample space for spades.



$SSS = \frac{1}{64}$

$SSS' = \frac{3}{64}$

$SS'S \quad \frac{3}{64}$

$SS'S' \quad \frac{9}{64}$

$S'SS \quad \frac{3}{64}$

$S'SS' \quad \frac{9}{64}$

$S'S'S \quad \frac{9}{64}$

$S'S'S' \quad \frac{27}{64}$

b) Find the probability that all 2 cards are spades.

$\frac{1}{64}$

c) Find the probability of getting at most 1 spade.

$\frac{27}{64}$

$SS'S'$

$S'SS'$

$S'S'S$

$\frac{9}{64} + \frac{9}{64} + \frac{9}{64}$

Question 4	9 marks (5, 1, 1, 1, 1)

From the 20 people in an aquatic team, 15 like swimming (S), eight like diving (D) and three like both swimming and diving.

a. Represent the information in a two-way table.

	S	**S'**	**T**
D	3	5	8
D'	12	0	12
T	15	5	20

b. Using your table, state the number of people who:

i don't like diving.

12

ii like only diving.

5

c. A person from the aquatic team is chosen at random. Find the probability that the person:

i likes both swimming and diving.

$\frac{3}{20}$

ii likes only diving.

$\frac{5}{20} = \frac{1}{4}$

Question 4	9 marks (5, 1, 1, 1, 1)

From the 20 people in an aquatic team, 15 like swimming (S), eight like diving (D) and three like both swimming and diving.

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D	3	5	8
D'	12	0	12
T	15	5	20

b. Using your table, state the number of people who:

i don't like diving.

12

ii like only diving.

5

c. A person from the aquatic team is chosen at random. Find the probability that the person:

i likes both swimming and diving.

$\frac{3}{20}$

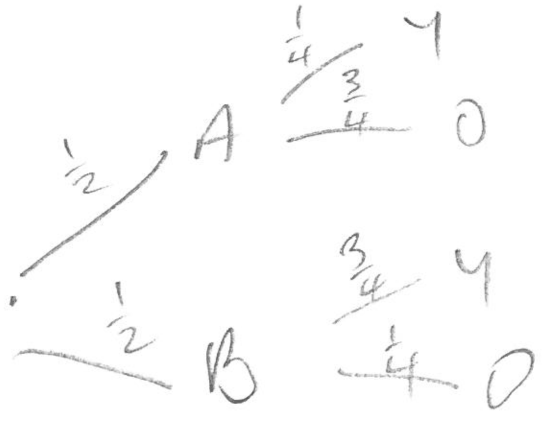
ii likes only diving.

$\frac{5}{20} = \frac{1}{4}$

Question 5	9 marks (3, 2, 2, 2)

Room A and Room B contain 4 chairs each. Room A contains 1 yellow chair and 3 orange chairs and Room B contains 3 yellow chairs and 1 orange chair. A Room is chosen at random and then a single chair is selected.

a. Represent the options available by drawing a tree diagram



A Y $\frac{1}{8}$

A O $\frac{3}{8}$

B Y $\frac{3}{8}$

B O $\frac{1}{8}$

b. If Room A is chosen, what is the probability of selecting a yellow chair?

$$\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$$

c. What is the probability of selecting Room B and a yellow chair?

$$\frac{3}{8} \left(\frac{3}{4} \times \frac{1}{2} \right)$$

d. What is the probability of selecting an orange chair?

$$\frac{1}{8} \times \frac{3}{8} = \frac{4}{8}$$

Question 6	**13 marks (2, 2, 1, 1,1, 2, 2, 1, 1)**

Two letters are chosen from the word PIG.

a) Complete a two-way table listing the sample space if selections are made:

i. with replacement

	P	I	G
P	PP	PI	PG
I	PP	II	IG
G	GP	GI	GG

ii. without replacement

	P	I	G
P	X	PI	PG
I	IP	X	IG
G	GP	GI	X

b) State the total numbers of outcomes if selection is made:

i. with replacement

9

ii. without replacement

6

c) If selection is made with replacement, find the probability that:

i. the two letters are the same

$\frac{3}{9} = \frac{1}{3}$

ii. there is a P or a G

$\frac{8}{9}$

v. there is a P and a G

$\frac{2}{9}$

d) If selection is made without replacement, find the probability that:

i. there is not an I

$\frac{2}{6}$

iii. there is at least a P

$\frac{4}{6}$